



Online Billing System

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Problem Statement

- Manual billing is time-consuming and error-prone
- Difficulty managing customer records & invoices
- Lack of automated payment tracking
- Need for digital billing and reporting

Objectives

- Automate billing and invoice generation
- Reduce calculation errors significantly
- Maintain customer and transaction records
- Generate detailed reports efficiently

Technologies Used

Backend: Node.js & Express.js

Database: MongoDB

Frontend: HTML, CSS, JavaScript

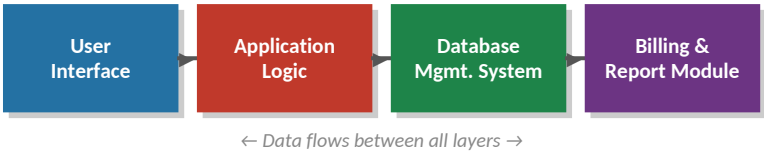
Framework: React

Tools: VS Code & Trae

Proposed System — Workflow



System Architecture



Implementation — UI Mockups

Invoice Generation — Live Screenshot

OBS					
Dashboard Customers Products Invoices Settings Logout					
Invoices + Create New Invoice					
Invoice #	Date	Customer	Amount	Status	
INV-20260206-8432	6/2/2026	Farhan	\$10000.00	Paid	
INV-20260205-2500	5/2/2026	Farhan Vohra	\$1260.00	Unpaid	
INV-20260205-2754	5/2/2026	Farhan	\$10500.00	Unpaid	
INV-20260205-2933	5/2/2026	Farhan	\$1070.00	Unpaid	
INV-20260204-4041	4/2/2026	Farhan	\$104.00	Unpaid	

Key Features

- Auto Invoice**
Instant PDF invoice generation
- Analytics**
Real-time billing reports & charts
- Search & Filter**
Quick customer & record lookup
- Secure Auth**
JWT-based login & role management
- Email Alerts**
Automated payment notifications
- Cloud Ready**
Scalable MongoDB Atlas support

Future Scope

- Online payment gateway integration (Razorpay/Stripe)
- Mobile application version (React Native)
- AI-powered billing anomaly detection
- Cloud deployment with CI/CD pipeline

References

- [1] O'Brien, J. A. (2010). Management Information Systems. McGraw-Hill.
- [2] Silberschatz, A. et al. (2019). Database System Concepts. McGraw-Hill.
- [3] MDN Web Docs. (2024). JavaScript & Web Technologies. developer.mozilla.org.
- [4] MongoDB Documentation. (2024). MongoDB Atlas & Node.js Driver. mongodb.com.

Conclusion

The Online Billing System provides a modern, efficient, and accurate digital solution for managing billing operations. Built with a MERN-inspired stack (Node.js, Express, MongoDB, React), the system eliminates manual billing errors and streamlines invoice generation, customer management, and payment tracking. It significantly improves operational efficiency and provides businesses with real-time insights through automated reporting — paving the way for a fully digitized billing ecosystem.